

Abstract No.

Co-targeting gangliosides to overcome antigen heterogeneity in H3-wildtype pediatric high-grade glioma

Kevin Lu^{1,2,*}, Lara Engertsberger¹, Guillermo Dalton¹, Won-Ju Kim¹, Luis Nagano¹, Carlos A.O. Biagi Junior¹, Alessandro Gasparetto³, Haley Ohlson³, Phuc Nguyen³, Alan MacKay⁴, Yura Grabovska⁴, Li Jiang¹, Costanza Lo Cascio¹, Joana Marques¹, Sina Neyazi¹, Alberto Bueno Costa¹, Stefano Testa¹, Anoushka Vattamparambil¹, Keith Ligon⁵, Chris Jones⁴, Roberto Chiarle³, Robbie Majzner¹, and Mariella Filbin¹

¹Dana-Farber Boston Children's Cancer and Blood Disorders Center, Boston

²Broad Institute of Harvard and MIT, Cambridge

³Boston Children's Hospital, Boston, MA

⁴Institute of Cancer Research, London, UK

⁵Dana-Farber Cancer Institute, Boston

*Presenting author

Pediatric brain tumors remain the leading cause of cancer-related death in children, with poor outcomes driven largely by high-grade glioma (HGG), for which 5-year survival remains below 20% despite standard-of-care multimodal therapy. We previously demonstrated promising clinical activity of GD2-directed CAR T cells in diffuse midline glioma (DMG) harboring H3K27M alterations, where the histone H3 mutation drives high, uniform GD2 expression. In H3 wildtype (H3wt) HGG, however, GD2 expression is more heterogeneous, limiting patient eligibility for current GD2-targeted approaches. We therefore sought to identify complementary ganglioside targets in H3wt HGG.

Analysis of bulk RNA-seq data from 624 HGG samples identified heterogeneous expression of *B4GALNT1*, the enzyme that converts precursor GD3 to GD2, as a key bottleneck for GD2 abundance in H3wt tumors. Immunohistochemistry in an independent cohort (n=20) confirmed marked spatial heterogeneity of *B4GALNT1* protein expression. Using both immunohistochemistry and flow cytometry on patient samples, we found that reduced *B4GALNT1* expression in GD2-low tumors was associated with compensatory accumulation of GD3.

To investigate the underlying biology, we correlated GD2 and *B4GALNT1* levels with transcriptomic programs in bulk and single-cell RNA-seq datasets from primary H3wt HGG patients. Higher *B4GALNT1*/GD2 expression was associated with a neuronal progenitor-like cell state, whereas lower *B4GALNT1* expression correlated with a mesenchymal program.

To overcome GD2 heterogeneity, we engineered and optimized a bispecific CAR T-cell construct capable of recognizing and eliminating tumor cells expressing either GD2 or GD3. Both antigens have been clinically targeted with favorable safety signals, and staining of normal tissues revealed no major overlapping expression. Together, these findings support dual GD2/GD3 targeting as a strategy to overcome antigen heterogeneity in H3wt HGG and expand ganglioside-directed CAR T-cell therapy to additional pediatric brain tumor patients.



Supported by SciSerTec

Web: <http://www.scisertec.de>